

S-GRS

Introduction

The Simplified Gravitational Reference Sensor (S-GRS) is an ultra-precise electrostatic inertial sensor for future Earth geodesy missions following the successful CHAMP [1], GRACE [2], GOCE [3], and GRACE-FO [4] missions. It is designed to measure non-gravitational forces and enhance the precision of gravity field recovery.

CMMT

The Charge Measurement and Metrology Testbed (CMMT) is a self-contained ground test apparatus which houses a mockup of an S-GRS sensor. The CMMT is intended to provide a platform to assess the long-period performance of the S-GRS inertial sensor stability and charge management system.

The duties of the CMMT are:

- Emulate the orbit flight environment pressure and temperature conditions (Figure 3)
- Isolate and anchor the Test Mass simulate (Figure 1)
- Provide controlled motion of the electrode housing with respect to the Test Mass simulate (Figure 3)
- Record environmental data, capacitive sensing data, laser interferometer stability, and residual charge on the Test Mass (Figure 2)

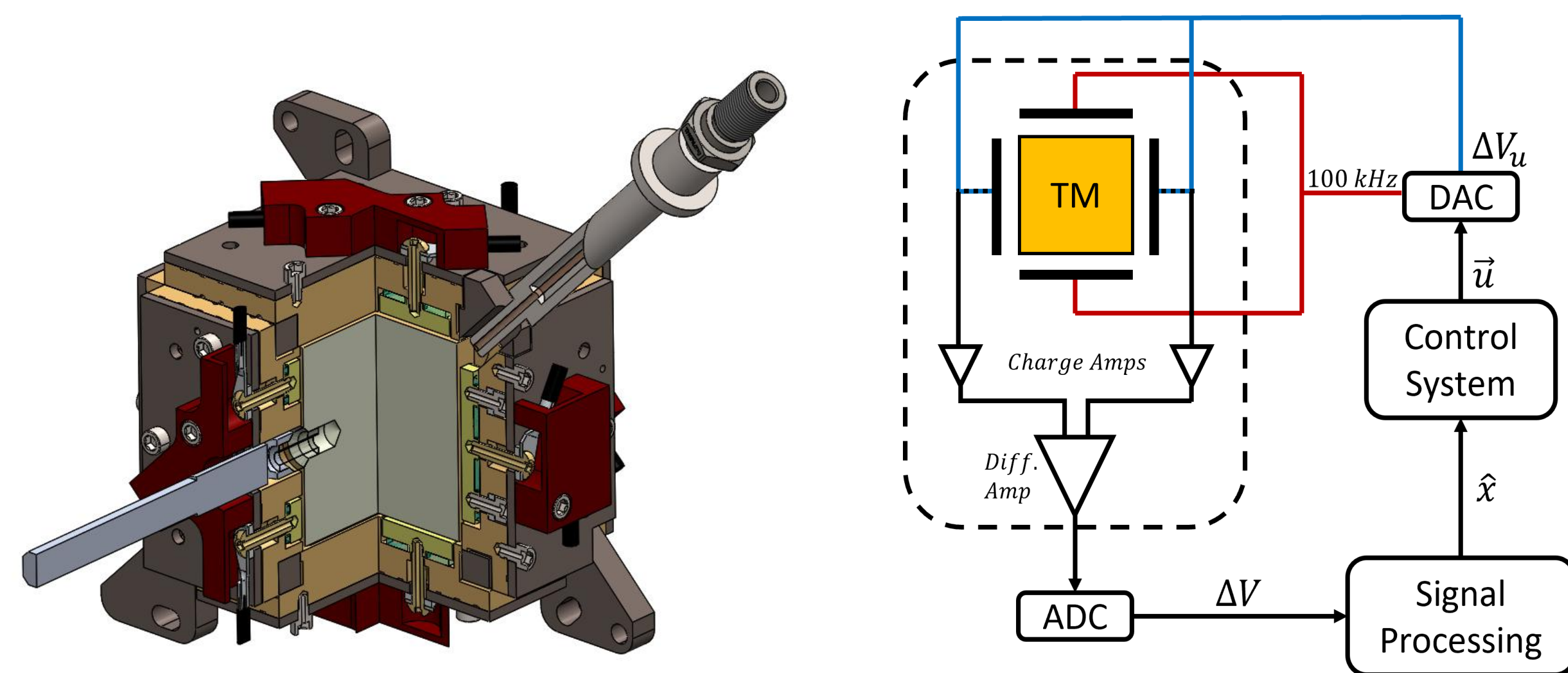


Figure 1: CMMT-modified S-GRS head internal view, showing anchoring scheme for AI Test Mass simulate

Figure 2: Sensing and actuation scheme of the S-GRS. Black traces represent capacitive data streams, blue represent actuation potential, and red represent injection signals.

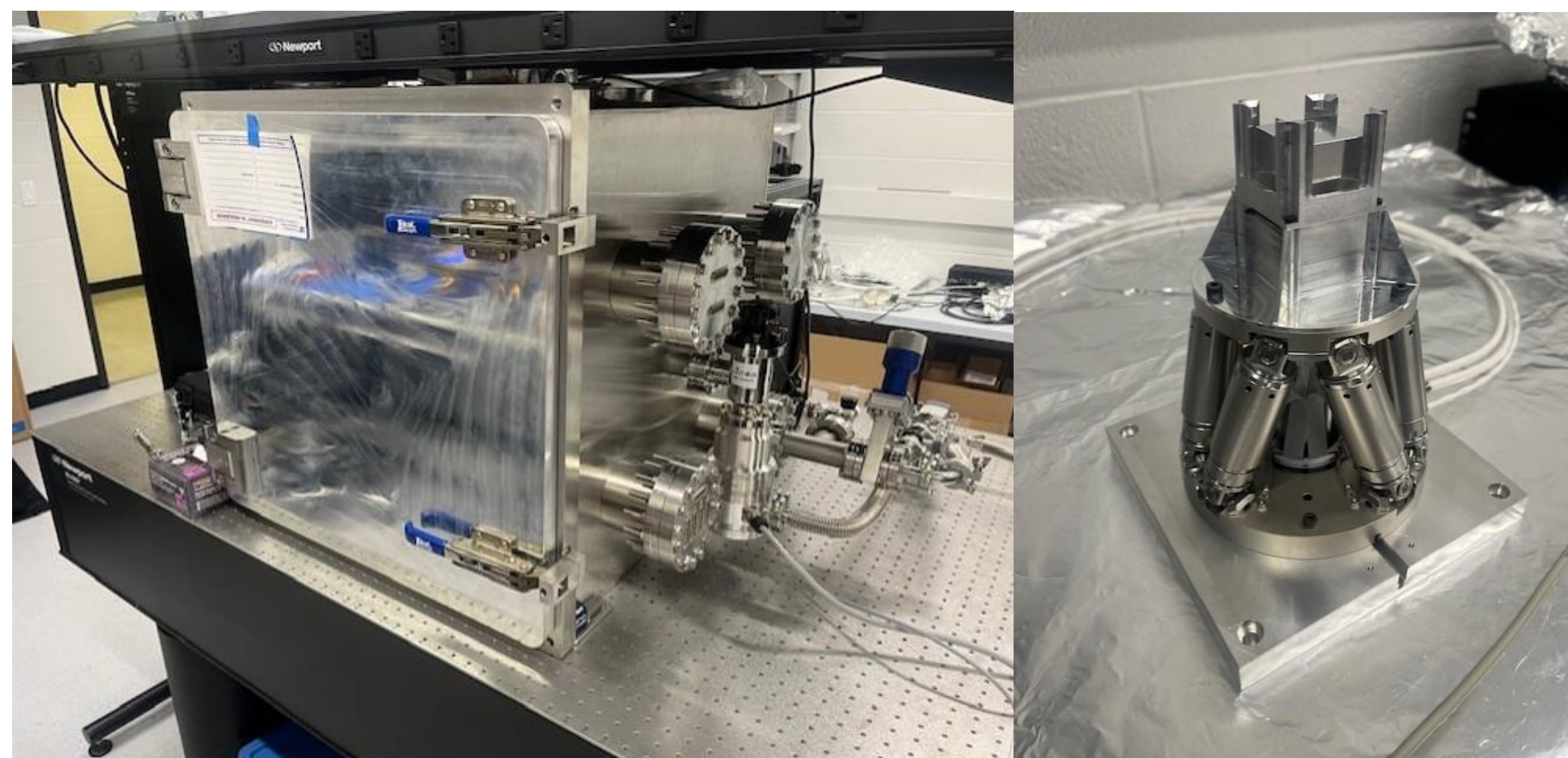


Figure 3: CMMT vacuum chamber and precision vacuum hexapod used to emulate the on-orbit flight environment. The superstructure on the hexapod is a surrogate S-GRS used to safely test and monitor motion schemes in atmosphere

Acknowledgements

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Test Mass Simulate

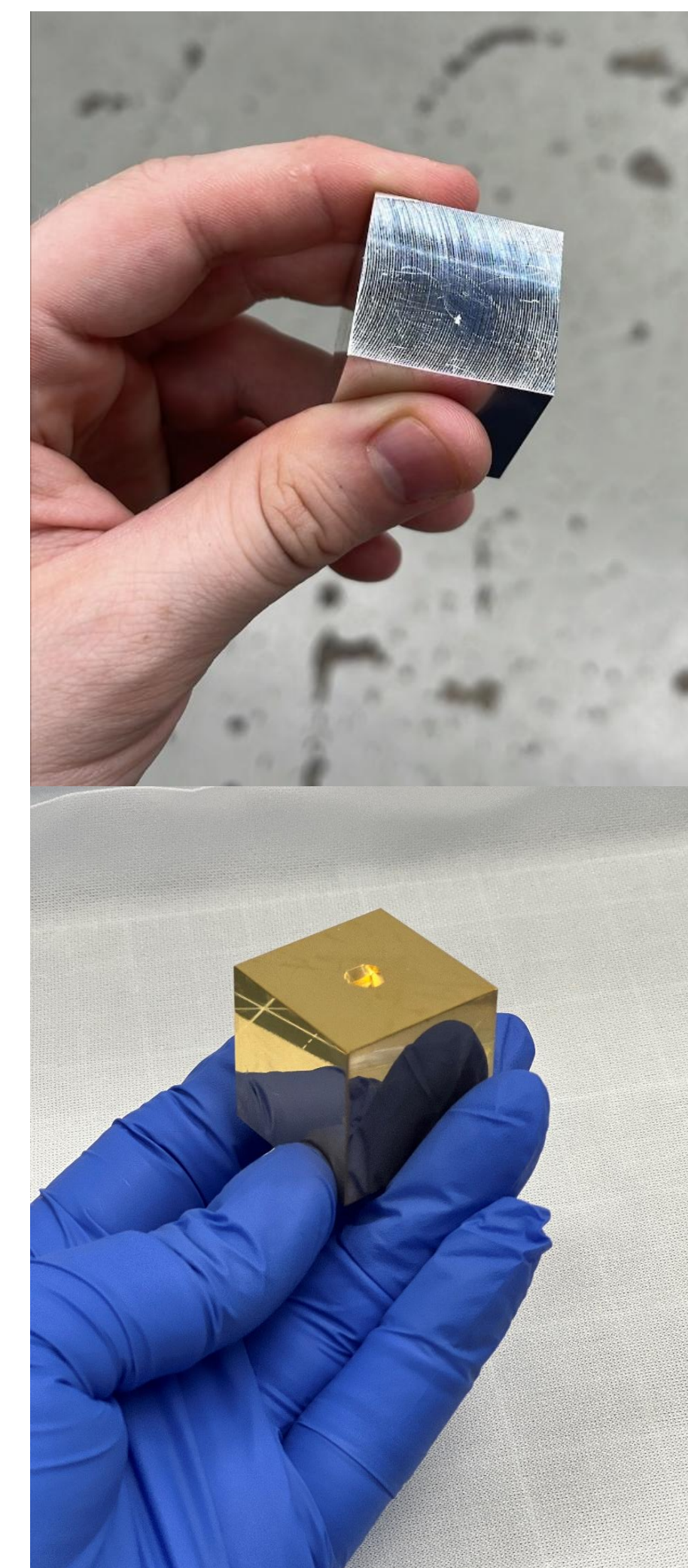


Figure 4: Uncoated test mass blank (top) and finished test mass simulate (bottom)

Tolerance Requirements

- Test mass is a 30 mm cube to maintain a 0.5 mm gap size,. Tolerances for each face affects gap size uncertainty.
- Aluminum test masses manufactured at PSSSL to within bilateral 25 μm tolerance.

Manufacturing

- CMMT test mass is aluminum instead of tungsten as mass isn't a concern and aluminum could be manufactured at UF.
- Grabbing finger and support shaft geometry were added for flight-similarity and ground testing compatibility.
- All test masses were gold coated for conductivity (Figure 4).

Metrology

- 8 test masses were manufactured within tolerance in-house.
- Average gap size variation of 2%, or 10 μm, across all face pairs for the 4 most precise test masses (Figure 5).
- Tolerances need to be confirmed with a coordinate measurement machine post gold coating before installation.

	X (1)	Y (2)	Z (3)
E	$\begin{bmatrix} 6 & 0 & 0 \\ 4 & \Delta & -4 \\ 1 & 0 & -5 \end{bmatrix}$	$\begin{bmatrix} -1 & -1 & 2 \\ 0 & 0 & 4 \\ 0 & 1 & 3 \end{bmatrix}$	$\begin{bmatrix} -8 & -7 & -4 \\ -7 & -7 & -5 \\ -2 & -6 & -4 \end{bmatrix}$
F	$\begin{bmatrix} 4 & 8 & 8 \\ -2 & \Delta & 2 \\ -7 & -2 & 2 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 \\ -3 & 1 & -5 \\ -6 & -6 & -7 \end{bmatrix}$	$\begin{bmatrix} -5 & -2 & 3 \\ -4 & 0 & 4 \\ -2 & 1 & 6 \end{bmatrix}$
B	$\begin{bmatrix} -6 & -5 & -1 \\ -6 & \Delta & -1 \\ -6 & -6 & -1 \end{bmatrix}$	$\begin{bmatrix} -1 & -9 & -15 \\ 2 & -6 & -11 \\ 5 & -3 & -9 \end{bmatrix}$	$\begin{bmatrix} 6 & 2 & -1 \\ 4 & -3 & -5 \\ 1 & -5 & -8 \end{bmatrix}$
H	$\begin{bmatrix} -5 & 3 & -1 \\ 3 & \Delta & 11 \\ 0 & 1 & 4 \end{bmatrix}$	$\begin{bmatrix} 8 & 4 & -8 \\ 4 & 5 & -2 \\ 3 & 5 & -2 \end{bmatrix}$	$\begin{bmatrix} 2 & -1 & -6 \\ 7 & 4 & 0 \\ 11 & 8 & 6 \end{bmatrix}$

Figure 5: Metrological report for 4 test mass samples showing deviation in μm from the nominal 30 mm side length

Capacitive Sensing Analog Electronics

Requirements

- Analog signal sensing has a noise requirement of 0.7 nm/√Hz
- New boards with dev components removed are being tested to re-confirm performance.

Performance

- <1 nm/√Hz above 50 mHz. Noise in lower frequencies is due to the unregulated thermal environment. Testing in a thermally controlled environment is expected to improve performance.
- Newest iteration includes anti-aliasing filter for further improved performance.

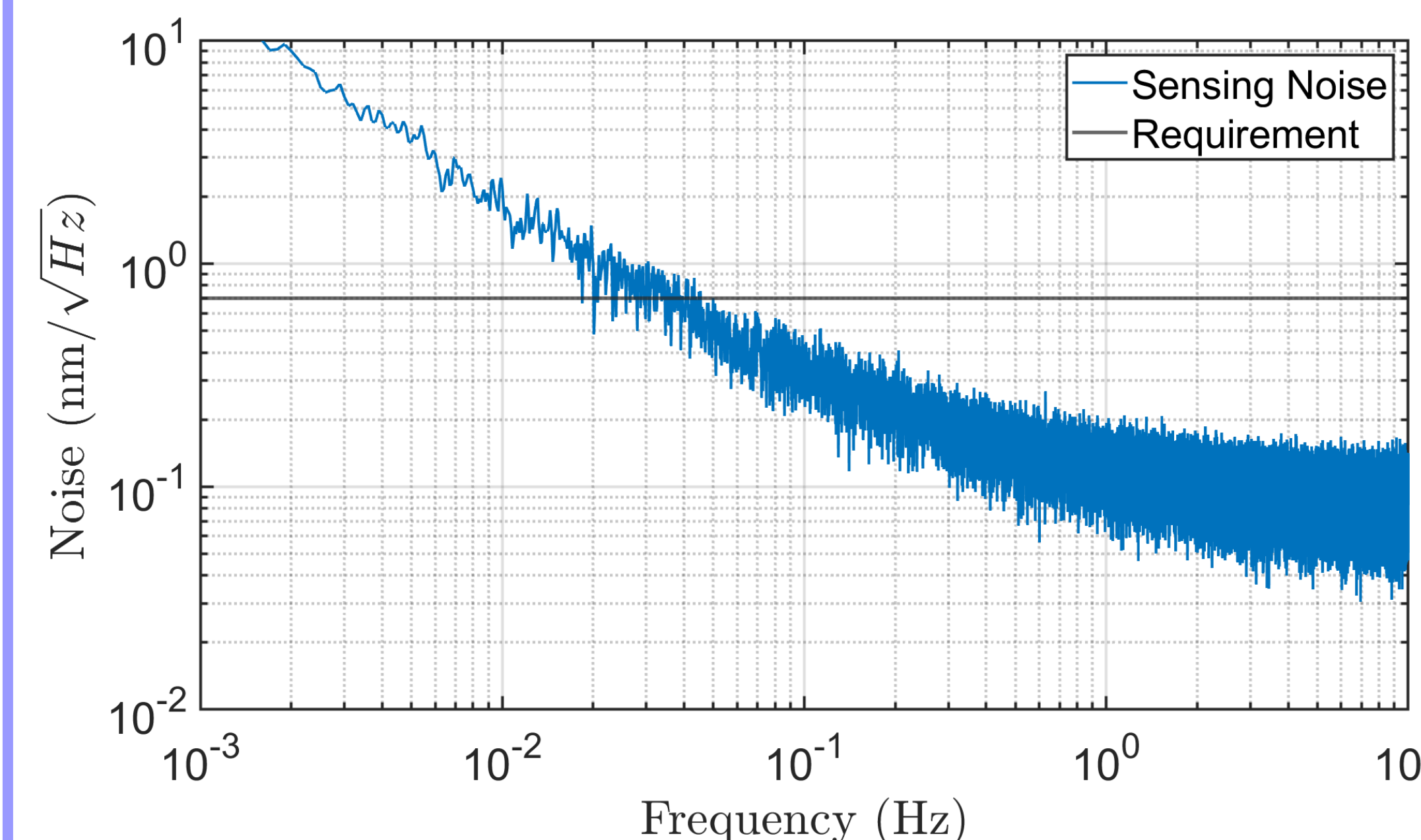


Figure 6: Power spectral density of the DC signal produced from running the sensing board output through IQ demodulation and an ADC.

UV Charge Management System

UV Injection System

- Serves to remotely control the Test Mass charge via the photoelectric effect
- Replaces grounding wire used in previous accelerometers
- Light is cast through an aperture in the electrode housing frame, guided by a fiber optic ferrule

Performance

- Measured 10% light transmittance from UV LED source into ferrule
- Light source generated from prototype unit intended for use on LISA

CMMT Test Mass Charge Measurement

- Charge accumulation on Test Mass measured via electrometer
- Contact made through grabbing figure aperture, interfacing with triangular indent
- Allows for capacitive sensing measurements with grabbing finger analog

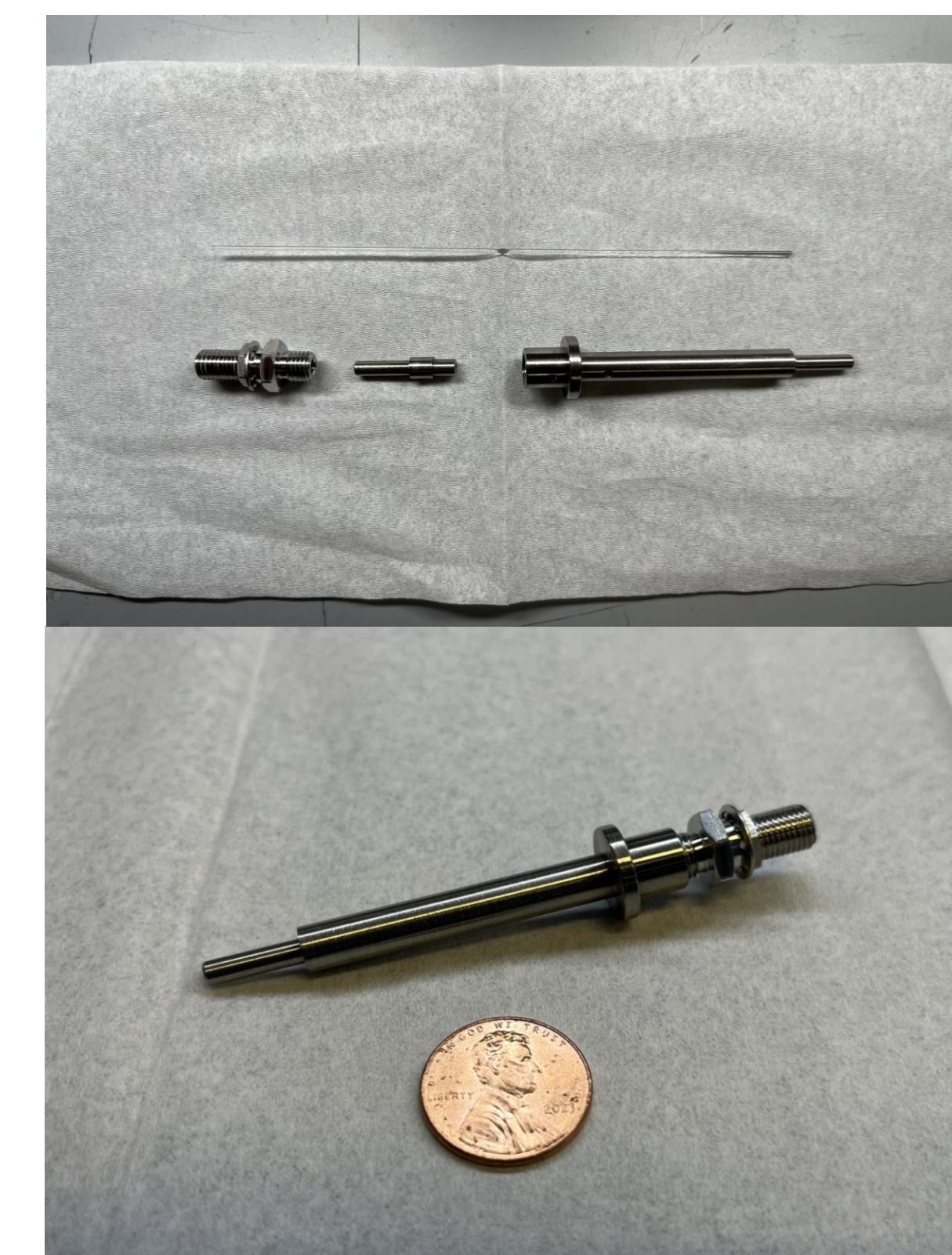


Figure 7: Titanium UV injection ferrule example showing the UV fiber adapter, alignment pin, and spindle (top), and the assembled ferrule with penny for scale (bottom)

Laser Ranging Interferometer



Figure 8: LRI hardware showing the vacuum-side optics including the science and reference laser tracks and TMA, and the atmosphere-side optics with photodetectors

Laser Ranging Interferometer

- Independent corroboratory measurement for position readout
- Dual-path heterodyne interferometer using a 1064 nm, 500 mW laser
- Utilizes a Triple Mirror Assembly (TMA) as a retroreflector for the LRI

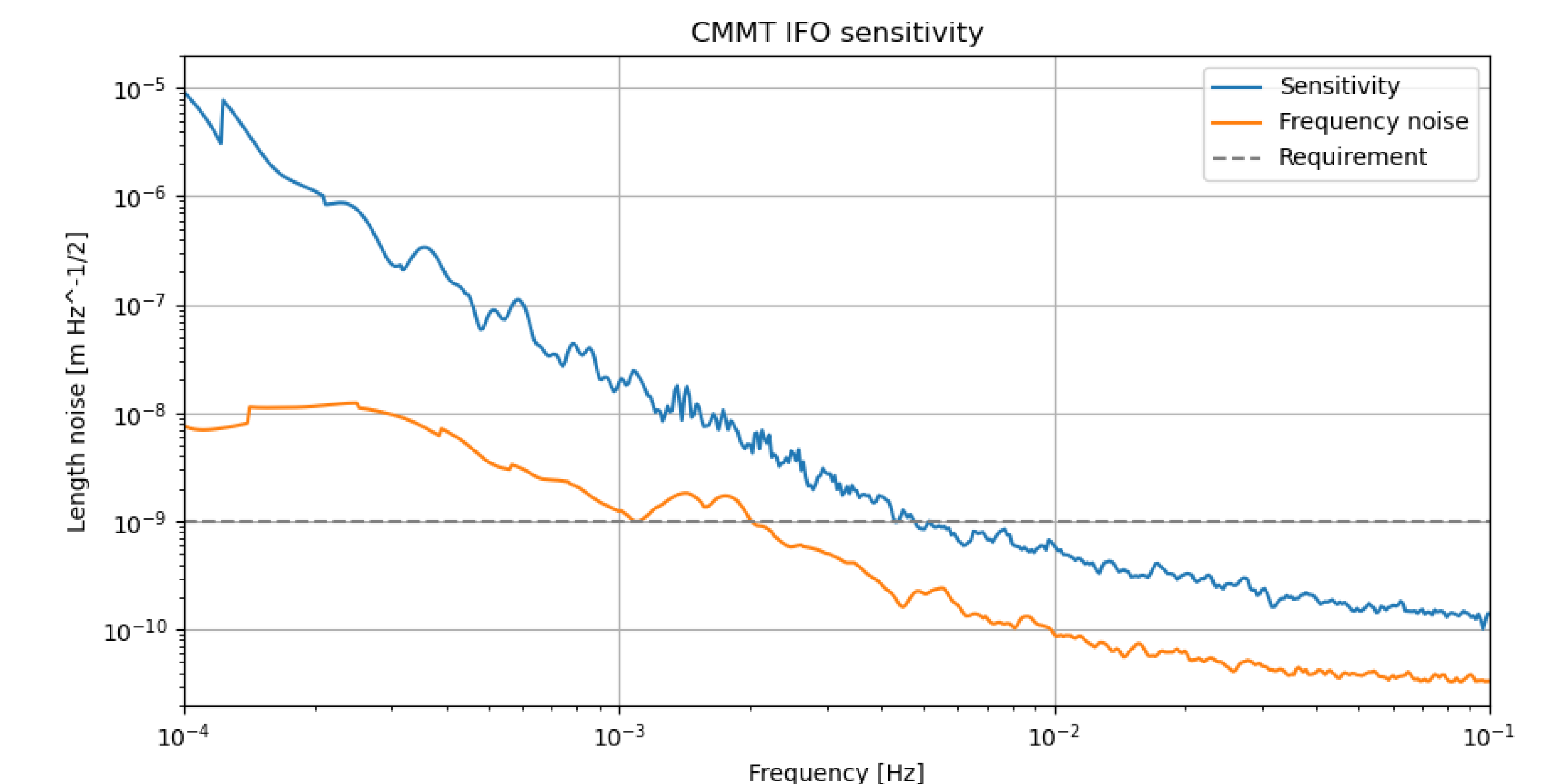


Figure 9: LRI path length noise spectra. Experiments performed in atmosphere and skewed by polarization noise and thermal fluctuations

Summary

- The CMMT will test the position noise stability of the S-GRS in a flight-like environment
- Test masses were manufactured in-house at UF to within a 25 μm tolerance, have been gold-coated, and are ready for use.
- Analog sensing noise is close to the requirement, with improvements expected in a controlled thermal environment.
- UV Injection for charge management uses proven technology and should be expected to effectively discharge the Test Mass
- The embedded LRI system is currently limited by polarization noise introduced by its retroreflector, intended to be addressed with a thin-film polarizer added to its center mirror

References

- Reigber C, Lühr H, Schwintzer P (2002)
- Tapley et al 2004
- Drinkwater et al 2003
- Sheard et al 2012
- A Simplified Gravitational Reference Sensor for Satellite Geodesy